

AlgoRecall

Adaptive DSA Revision and Analytics Platform

Project Proposal for UCS503P

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1. Abstract

AlgoRecall will be a web-based platform designed to help users manage and organize the revision of previously solved DSA questions. The system will allow users to store solved questions along with details such as topic, difficulty level, confidence level, and revision status.

The platform will maintain a complete revision history and schedule future revisions based on user-defined and system-generated revision parameters. It will provide a dedicated dashboard for viewing due questions and tracking revision activities. Users will be able to search, filter, and categorize questions to efficiently manage their revision workflow.

In addition, the system will offer analytical reports and visualizations that will summarize revision progress, topic-wise performance, confidence distribution, and revision consistency.

2. Introduction

Data Structures and Algorithms (DSA) play a crucial role in technical interviews and competitive programming. As part of their preparation, students and job aspirants solve a large number of problems across various topics such as Arrays, Linked Lists, Trees, Graphs, Dynamic Programming, and Greedy Algorithms. Continuous practice helps learners develop problem-solving skills and understand different algorithmic patterns.

As the number of solved questions grows, managing and revisiting previously learned concepts becomes increasingly important. Learners often maintain records of solved problems using spreadsheets, notes, or online coding platforms. However, these methods primarily focus on problem-solving activities and provide limited support for organizing and tracking revisions.

To address this need, AlgoRecall is proposed as a web-based platform that will enable users to maintain a structured record of solved questions, track revision activities, and monitor their overall learning progress through various management and analytical features.

3. Problem Statement

Existing coding platforms and online judges primarily focus on helping users solve programming problems and evaluate their coding skills. While these platforms provide features such as problem categorization, progress tracking, and submission history, they offer limited support for systematic revision and long-term knowledge retention.

As learners solve an increasing number of DSA questions, it becomes difficult to remember previously learned concepts, problem-solving patterns, and algorithmic approaches. Revisions are often performed irregularly and without a structured plan, making it challenging for learners to determine which questions require more attention. Additionally, identifying weak topics and monitoring revision progress becomes increasingly difficult as the volume of solved questions grows.

Therefore, there is a need for a dedicated revision management system that can organize solved questions, schedule revisions effectively, maintain revision records, and provide meaningful insights into a learner's strengths and weaknesses. Such a system can support a more disciplined and structured approach to DSA revision and learning.

4. Objectives

The primary objective of AlgoRecall is to provide a structured platform for managing and revising previously solved DSA questions. The system will aim to support users in maintaining consistent revision habits and tracking their learning progress effectively.

The objectives of the project are:

- To store and manage records of solved DSA questions.
- To allow users to assign and update confidence levels for each question.
- To schedule and organize future revisions based on revision-related parameters.
- To maintain a complete history of revision activities.
- To identify weak topics and areas that require additional practice.
- To support a structured and disciplined revision workflow.
- To improve long-term retention of DSA concepts and problem-solving patterns.

5. Scope of the Project

In Scope

- a. User Registration and Authentication
- b. Question Management (Add, Edit, Delete, View)
- c. Confidence Level Tracking
- d. Revision Scheduling
- e. Revision History
- f. Analytics and Reports
- g. Search and Filter Functionality

Out of Scope

- h. Online Judge Functionality
- i. Code Compilation and Execution
- j. Contest Management
- k. AI-based Solution Generation
- l. Automatic Leet-code Synchronization
- m. Real-time Collaboration Features

The project will be focused on revision management and learning analytics rather than problem solving or code execution.

6. Target Users

The primary stakeholders of AlgoRecall shall be DSA learners, students preparing for technical interviews, and competitive programmers.

7. Functional Requirements

The system will provide the following functionalities:

- a. User Registration and Login
- b. Add, Edit, and Delete Questions
- c. Revise Questions and Record Revision Activity

- d. Update Confidence Levels
- e. View Due Questions for Revision
- f. View Analytics and Progress Reports
- g. Search and Filter Questions

8. Non-Functional Requirements

The system will satisfy the following quality requirements:

- a. **Performance:** Quick response time for user operations.
- b. **Security:** Secure authentication and protection of user data.
- c. **Usability:** Simple and user-friendly interface.
- d. **Scalability:** Ability to handle increasing amounts of data and users.
- e. **Maintainability:** Easy to update and extend in the future.
- f. **Availability:** Reliable access to the system whenever required.

9. System Features

The system will consist of the following major modules:

- a. **Authentication:** User registration, login, and secure access management.
- b. **Question Management:** Add, edit, delete, and organize solved DSA questions.
- c. **Revision Engine:** Schedule and manage future revisions.
- d. **Due Today Dashboard:** Display questions that are due for revision.
- e. **Revision History:** Maintain records of past revision activities.
- f. **Analytics:** Provide insights into revision progress and performance.
- g. **Search & Filter:** Enable quick retrieval of questions based on different criteria.

10. Database Design

The system will use a MongoDB database consisting of three primary collections:

- a. **Users:** User information such as name, email, password hash, and account-related details required for authentication and authorization.
- b. **UserQuestions:** All questions added by a user along with attributes such as title, platform, topic, difficulty level, confidence level, date solved, last revision date, next revision date, revision count, and current revision status.
- c. **RevisionHistory:** Records of individual revision sessions, including the associated question, revision date, confidence updates, and other revision-related information.

11. Revision Engine

The Revision Engine will be responsible for scheduling and managing question revisions. It will utilize factors such as **Confidence Level**, **Question Difficulty**, and **Last Revision Date** to determine when a question should be revised next.

Based on these parameters, the system shall calculate the next revision date and assign a priority to each question. Questions with lower confidence levels, higher difficulty, or longer gaps since the last revision shall be given higher priority.

12. User Workflow

The typical workflow of the system will be as follows:

1. The user registers or logs into the platform.
2. Adds solved DSA questions with details such as topic, difficulty, and confidence level.
3. The system schedules future revisions for the added questions.
4. The user views questions that are due for revision through the dashboard.
5. The user revises a question and updates their confidence level.
6. The system records the revision activity and reschedules the next revision.
7. The user can monitor progress and performance through analytics and reports.

13. Technical Stack to be Used:

Frontend Layer

- React.js
- Tailwind CSS
- JavaScript (ES6+)

Backend Layer

- Node.js
- Express.js
- RESTful APIs

Database Layer

- MongoDB Atlas
- Mongoose ODM
- Authentication & Security
- JSON Web Tokens (JWT)
- Password Hashing (bcrypt)

Data Visualization

- Chart.js

Version Control & Development Tools

- Git
- GitHub
- Visual Studio Code

14. Expected Outcomes

The project is expected to improve revision consistency, retention of DSA concepts, identification of weak topics, and overall interview preparation.